

Practical No. 1

Title:- Program to demonstrate the features of Dart Language.

a) Features of DART Programming Language.

b) Introduction to Visual Studio Code.

c) How to install Flutter SDK.

Dart is an open-source general-purpose programming language developed by Google. It supports application development in both client and server-side. But it is widely used for the development of android apps, iOS apps, IoT (Internet of Things), and web applications using the Flutter Framework.

a) Features of DART Programming Language:

The Dart programming language offers several key features:

- 1. Open Source:** Dart is freely available, developed by Google, and approved by the ECMA standard. It comes with a BSD license.
- 2. Platform Independent:** Dart runs on Windows, Linux, Macintosh, and other operating systems. It has its own Virtual Machine (Dart VM) for cross-platform execution.
- 3. Object-Oriented:** Dart supports OOP concepts like classes, inheritance, interfaces, and optional typing. It also includes advanced features like mixins, abstract classes, and a robust type system.
- 4. Concurrency:** Dart is asynchronous and supports multithreading using Isolates. Isolates communicate via message passing.
- 5. Extensive Libraries:** Dart provides built-in libraries (SDK, core, math, async, etc.) and allows organizing code into reusable libraries.
- 6. Easy to Learn:** If you know languages like Java, C#, or JavaScript, learning Dart is straightforward due to its similar syntax.
- 7. Flexible Compilation:** Dart supports both Ahead-of-Time (AOT) and Just-in-Time (JIT) compilation.
- 8. Type Safe:** Dart uses static type checking and runtime checks, making code more readable and reliable.
- 9. Treats Everything as an Object:** In Dart, everything is treated as an object.

b) Introduction to Visual Studio Code:

Visual Studio Code, often referred to as VS Code, is a powerful source-code editor developed by Microsoft. It's available for Windows, Linux, macOS, and even as a web-based version. Here are some steps to get started with it:

1. Installation and Setup:

- **Download and Install:** You can download VS Code from the official website. (Install VSCodeUserSetup-x64_1.81.1)
- **Open a Folder:** After installation, open VS Code and use the menu option File > Open Folder (or press Ctrl+K Ctrl+O on Windows/Linux) to open a project folder.
- **Explorer View:** Access your files and folders using the Explorer view (shortcut: Ctrl+Shift+E on Windows/Linux).
- **Search View:** Use the Search view (shortcut: Ctrl+Shift+F on Windows/Linux) to find text across your project.
- **Source Control View:** Manage version control with Git using the Source Control view.

2. Editing and Customization:

- **Command Palette:** Access various commands and settings via the Command Palette (shortcut: Ctrl+Shift+P on Windows/Linux).
- **Extensions:** Enhance VS Code by installing extensions for different languages and frameworks (shortcut: Ctrl+Shift+X on Windows/Linux).
- **Themes:** Customize the editor's appearance by choosing a color theme.

3. Running and Debugging:

- **Integrated Terminal:** Use the integrated terminal to run commands and scripts.
- **Run and Debug:** Start debugging your code (shortcut: F5).
- **Programming Language Extensions:** Install extensions for languages like Python, JavaScript, Dart and more.
- VS Code is widely used by developers due to its flexibility, rich features, and active community.

c) How to install Flutter SDK:

Step 1:- Go to Browser → type flutter_windows_3.22.0-stable download → Install | FlutterInstall | Flutter → Click on Windows Current Device → Choose Android → Click Download and Install Tab → Download the zip file → After Downloading zip file right click on it and select option extract here → You will get the flutter folder.

Step 2:- Go to C drive → Create New Folder and name it as src → copy that flutter folder and paste it inside C drive src folder.

Step 3:- Go to C drive → src → flutter → bin → copy path (C:\src\flutter\bin) → Click on This PC → Properties → Advanced System Settings → Click on Environment Variables → Under User variables → Select Path variable → Edit → C:\src\flutter\bin (Paste Path here) → Again Click on New Button → C:\src\flutter
→ OK.

Step 4:- Go to VS Code → Extensions (Ctrl+Shift+X) → Type Flutter → click on Flutter Dart Code → Install.

Step 5:- Go to Explorer (Ctrl+Shift+E) → View Tab → Command Palette → type flutter → Flutter New Project → Locate Folder → Go to C drive → src → Select flutter folder → Click on Set flutter SDK folder.

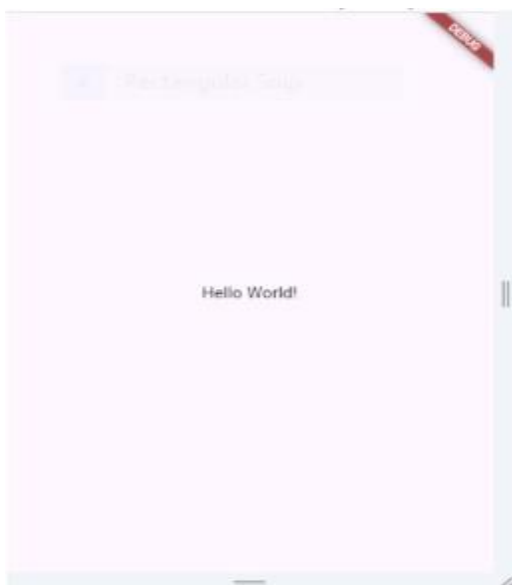
Step 6:- Create a folder on Desktop to save all files.

Step 7:- View Tab → Command Palette → type flutter → Flutter New Project → Empty Application → Select the folder to create folder in → Give proper name to application (Ex. Flutter_App_1) → Click on Checkbox → Yes, I trust the authors → After some installation you will get the main.dart file.

Step 8:- Right side of Lower Panel → Window(windows-x64) → Click on it → Select Chrome as browser.

Step 9:- To run an application press Ctrl+F5 (Run without Debugging).

Step 10:- Output for default “Hello World!” page (main.dart).



The `main()` function is a predefined method in Dart. This method acts as the entry point to the application. A Dart script needs the `main()` method for execution. `print()` is a predefined function that prints the specified string or value to the standard output i.e. the terminal.

Introduction to Dartpad

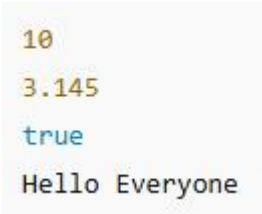
Go to browser → type dartpad → click on online editor link DartPad

1) Example for type of the variable using

Dartpad: Code:

```
void main() {  
    // Declaring and initialising variable  
    int num1 = 10;  
    double num2 = 3.145;  
    bool num3 = true;  
    String msg = 'Hello Everyone';  
    // Printing values of all variables  
    print(num1);  
    print(num2);  
    print(num3);  
    print(msg);  
}
```

Output:



```
10  
3.145  
true  
Hello Everyone
```

2) Example for Different types of Operators in**Dart: Code:**

```
void main()
{
    int a = 30, b = 20;

    // Addition
    var c = a + b;
    print('Sum of a and b is $c');

    // Subtraction
    var d = a - b;
    print('The difference between a and b is $d');

    // Using Unary minus
    var e = -d;
    print('The negation of difference is $e');

    // Multiplication
    var f = a * b;
    print('The multiplication of a and b is $f');

    // Division
    var g = a / b;
    print('The division of a and b is $g');

    // Quotient
    var h = a ~/ b;
    print('The quotient of a and b is $h');

    // Remainder
    var i = a % b;
    print('The remainder of a and b is $i');
}
```

Output:

```
Sum of a and b is 50
The difference between a and b is 10
The negation of difference is -10
The multiplication of a and b is 600
The division of a and b is 1.5
The quotient of a and b is 1
The remainder of a and b is 10
```

3) Example for Decision making**statement: Code:**

```
void main()
{
    var marks = 74;
    if (marks > 75)
    {
        print('Distinction');
    }
    else if (marks >= 60)
    {
        print('First Class');
    }
    else if (marks >= 40)
    {
        print('Second Class');
    }
    else
    {
        print('Fail');
    }
}
```

Output:

```
First Class
```

4) Example for**Function: Code:**

```
void main()
{
    print(factorial(6));
}

factorial(number)
{
    if (number <= 0)
    {
        return 1;
    }
    else
    {
        return (number * factorial(number - 1));
    }
}
```

Output:720**5) Example for Defining a****Class: Code:**

```
// Defining class
class Student {
    var studName;
    var studAge;
    var studRno;
    // Defining class function
```

```
showInfo() {  
    print('Student Name is : ${studName}');  
    print('Student Age is : ${studAge}');  
    print('Student Roll Number is : ${studRno}');  
}  
  
void main() {  
    // Creating object called stud  
    var stud = new Student();  
    stud.studName = 'Harsh';  
    stud.studAge = 25;  
    stud.studRno = 105;  
    // Accessing class function  
    stud.showInfo();  
}
```

Output:

```
Student Name is : Harsh  
Student Age is : 25  
Student Roll Number is : 105
```


Practical No. 2

Title: - Designing the mobile app to implement different widgets.

a) Basic Widgets

b) Material Design Widgets

c) Custom Widgets

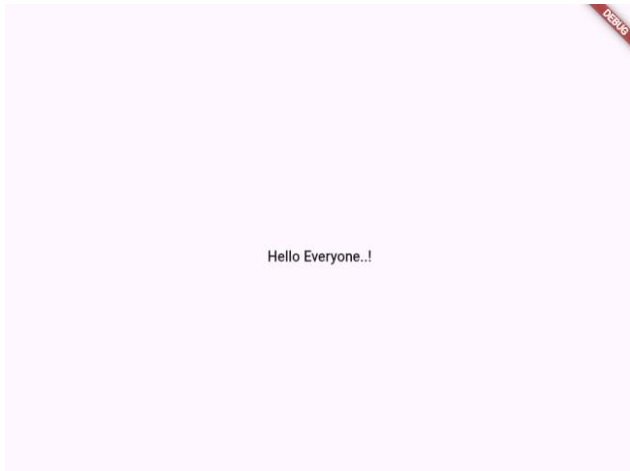
1) Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MainApp());
}

class MainApp extends StatelessWidget {
  const MainApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      home:
        Scaffold( body:
          Center( child:
            Text(
              'Hello Everyone..!',
              style: TextStyle(fontWeight: FontWeight.w900),
            ), // Text
          ), // Center
        ), // Scaffold
    ); // MaterialApp
  }
}
```

Output:**2) Input:**

```
import 'package:flutter/material.dart';

void main() => runApp(const ElevatedButtonExampleApp());

class ElevatedButtonExampleApp extends StatelessWidget
{ const ElevatedButtonExampleApp({super.key});

  @override
  Widget build(BuildContext context)
  { return MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: const Text('ElevatedButton Sample')),
      body: const ElevatedButtonExample(),
    ), ); } }

class ElevatedButtonExample extends StatefulWidget
{ const ElevatedButtonExample({super.key});

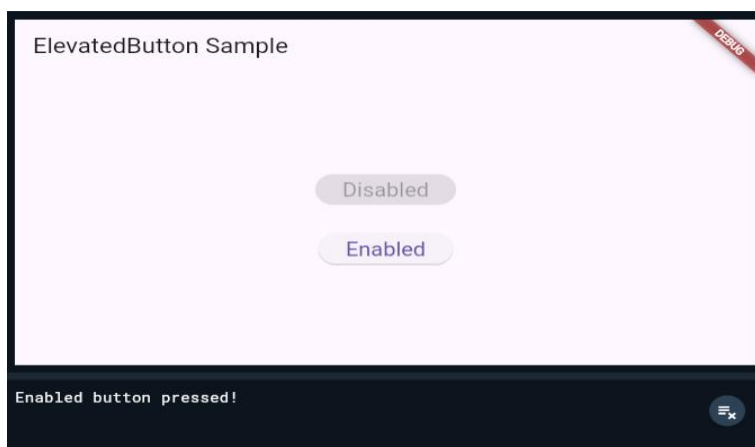
  @override
  State<ElevatedButtonExample> createState() => _ElevatedButtonExampleState();
}

class _ElevatedButtonExampleState extends State<ElevatedButtonExample>
{ final ButtonStyle style = ElevatedButton.styleFrom(
```

```

    textStyle: const TextStyle(fontSize: 20),
  );
  @override
  Widget build(BuildContext context)
  {
    return Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.min,
        children:
        <Widget>[ ElevatedButton(
          style: style,
          onPressed: null, // Disabled button
          child: const Text('Disabled'),
        ),
        const SizedBox(height: 30),
        ElevatedButton(
          style: style,
          onPressed: () {
            // You can add your onPressed logic here
            print('Enabled button pressed!');
          },
          child: const Text('Enabled'),
        ), ], ), ); } }

```

Output:

3) Input:

```

import 'package:flutter/material.dart';

// Flutter code sample for [AppBar].

void main() => runApp(const AppBarApp());

class AppBarApp extends StatelessWidget
{
  const AppBarApp({super.key});

  @override
  Widget build(BuildContext context)
  {
    return const MaterialApp(
      home: AppBarExample(),
    );
  }
}

class AppBarExample extends StatelessWidget
{
  const AppBarExample({super.key});

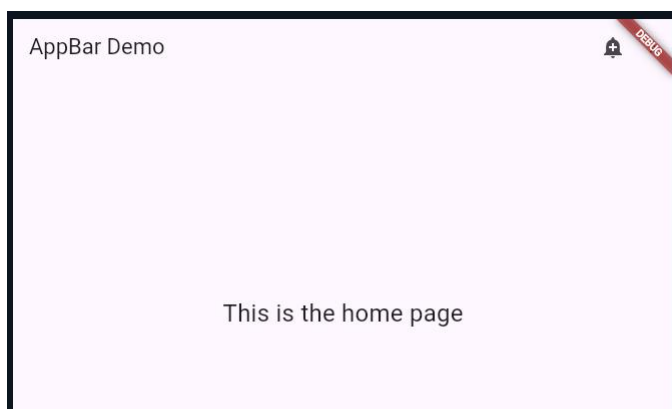
  @override
  Widget build(BuildContext context)
  {
    return Scaffold(
      appBar: AppBar(
        title: const Text('AppBar Demo'),
        actions: <Widget>[
          IconButton(
            icon: const Icon(Icons.add_alert),
            tooltip: 'Show Snackbar',
            onPressed: () {
              ScaffoldMessenger.of(context).showSnackBar(
                const SnackBar(
                  content: Text('This is a snackbar'),
                ),
              );
            },
          ),
          IconButton(
            icon: const Icon(Icons.navigate_next),
            tooltip: 'Go to the next page',

```

```

onPressed: ()
{ Navigator.push
( context,
MaterialPageRoute<void>( builder:
(BuildContext context) { return
Scaffold(
  appBar: AppBar(
    title: const Text('Next page'),
  ),
  body: const
    Center( child:
      Text(
        'This is the next page',
        style: TextStyle(fontSize: 24),
      ), ), ); }, ), ); }, ), ], ),
body: const
  Center( child: Text(
    'This is the home page',
    style: TextStyle(fontSize: 24),
  ), ), ); } }

```

Output:

4) Input:

```

import 'package:flutter/material.dart';

List<String> titles =
<String>[ 'Cloud',
  'Beach',
  'Sunny',
];

void main() => runApp(const AppBarApp());

class AppBarApp extends StatelessWidget
{ const AppBarApp({super.key});

  @override
  Widget build(BuildContext context)
  { return MaterialApp(
    theme: ThemeData(
      colorSchemeSeed: const Color(0xff6750a4),
      useMaterial3: true,
    ),
    home: const AppBarExample(),
  ); } }

class AppBarExample extends StatelessWidget
{ const AppBarExample({super.key});

  @override
  Widget build(BuildContext context) {
    final ColorScheme colorScheme = Theme.of(context).colorScheme;
    final Color oddItemColor = colorScheme.primary.withOpacity(0.05);
    final Color evenItemColor = colorScheme.primary.withOpacity(0.15);
    const int tabsCount = 3;

    return
      DefaultTabController( initi
        alIndex: 1,
        length: tabsCount,

```

child:

```
Scaffold( appBar:
AppBar(
  title: const Text('AppBar Sample'),
  notificationPredicate: (ScrollNotification notification)
    { return notification.depth == 1;
  },
  scrolledUnderElevation: 4.0,
  shadowColor: Theme.of(context).shadowColor,
  bottom: TabBar(
    tabs:
      <Widget>[ Ta
        b(
          icon: const Icon(Icons.cloud_outlined),
          text: titles[0],
        ),
        Tab(
          icon: const Icon(Icons.beach_access_sharp),
          text: titles[1],
        ),
        Tab(
          icon: const Icon(Icons.brightness_5_sharp),
          text: titles[2],
        ),
      ],
    ),
  ),
```

body:

```
TabBarView( childr
en:
  <Widget>[ ListVie
w.builder(
  itemCount: 25,
  itemBuilder: (BuildContext context, int index)
```

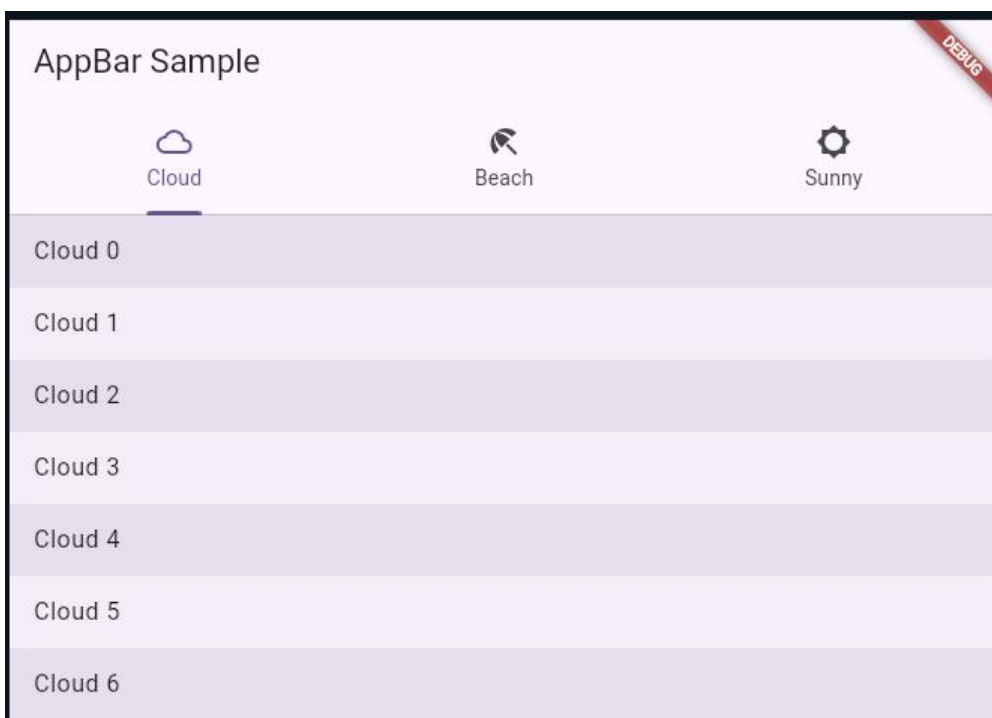
```

        { return ListTile(
          tileColor: index.isOdd ? oddItemColor : evenItemColor,
          title: Text('${titles[0]} $index'),
        ); },
      ),

ListView.builder(
  itemCount: 25,
  itemBuilder: (BuildContext context, int index)
    { return ListTile(
      tileColor: index.isOdd ? oddItemColor : evenItemColor,
      title: Text('${titles[1]} $index'),
    ); },
  ),

ListView.builder(
  itemCount: 25,
  itemBuilder: (BuildContext context, int index)
    { return ListTile(
      tileColor: index.isOdd ? oddItemColor : evenItemColor,
      title: Text('${titles[2]} $index'),
    ); },
  ), ],
),
),
); } }

```

Output:

5) Input:

```

import 'package:flutter/material.dart';

/// Flutter code sample for [Scaffold].

void main() => runApp(const ScaffoldExampleApp());

class ScaffoldExampleApp extends StatelessWidget
{
  const ScaffoldExampleApp({super.key});

  @override
  Widget build(BuildContext context)
  {
    return const MaterialApp(
      home: ScaffoldExample(),
    );
  }
}

class ScaffoldExample extends StatefulWidget
{
  const ScaffoldExample({super.key});

  @override
  State<ScaffoldExample> createState() => _ScaffoldExampleState();
}

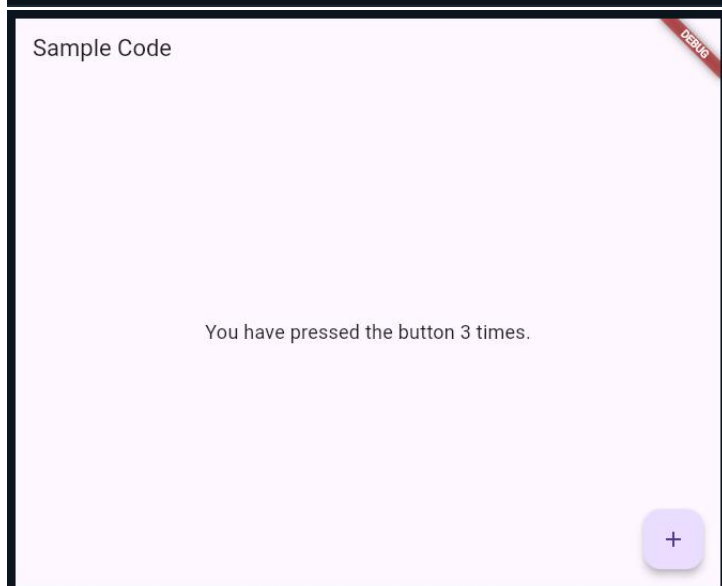
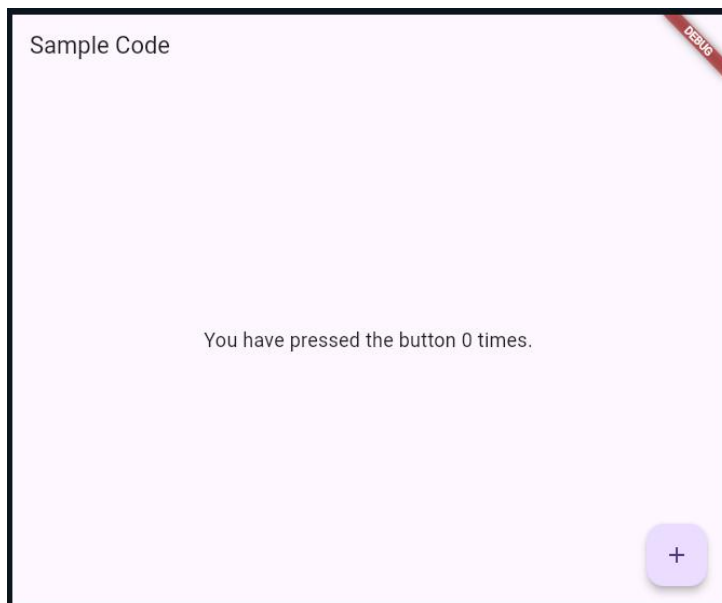
class _ScaffoldExampleState extends State<ScaffoldExample>
{
  int _count = 0;

  void _incrementCounter()
  {
    setState(() {
      _count++;
    });
  }

  @override
  Widget build(BuildContext context)
  {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Sample Code'),
      ),
      body:
        Center(
          child: Text(

```

```
'You have pressed the button $_count times.',  
style: const TextStyle(fontSize: 18),  
) , ),  
floatingActionButton:  
FloatingActionButton( onPressed:  
  _incrementCounter,  
  tooltip: 'Increment Counter',  
  child: const Icon(Icons.add),  
) , ); } }
```

Output:

Practical No. 3

Title: - Designing the mobile app to implement different Layouts.

1) Input:

```
import 'package:flutter/material.dart';

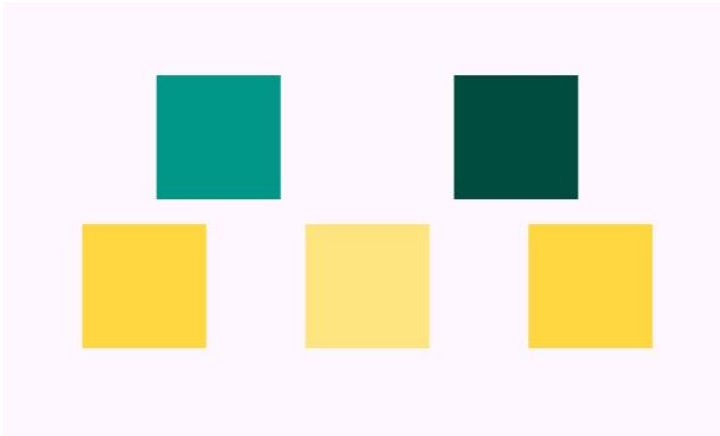
void main() {
  runApp(const DemoApp());
}

class DemoApp extends StatelessWidget
{
  const DemoApp({super.key});

  @override
  Widget build(BuildContext context)
  {
    return MaterialApp(
      title: 'My Application',
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        body: Padding(
          padding: const EdgeInsets.all(20.0),
          child: Center(
            child: Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                // First Row
                Row(
                  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
                  children: [
                    Container( height: 100,
                      width: 100,
                      color: Colors.teal,
```

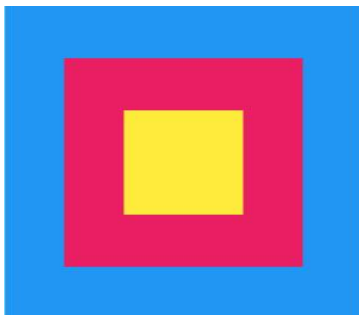
```

    ),
    Container( h
      eight: 100,
      width: 100,
      color: Colors.teal[900], // not const
    ), ], ),
const SizedBox(height: 20), // spacing between rows
// Second Row
Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    Container( h
      eight: 100,
      width: 100,
      color: Colors.amberAccent,
    ),
    Container( h
      eight: 100,
      width: 100,
      color: Colors.amberAccent[100],
    ),
    Container( h
      eight: 100,
      width: 100,
      color: Colors.amberAccent[200],
    ), ], ), ], ), ), ); } }
```

Output:**2) Input:**

```
import 'package:flutter/material.dart';
void main() => runApp(MyApp());
class MyApp extends StatelessWidget
{ @override
  Widget build(BuildContext context)
  { return MaterialApp(
    title: 'Multiple Layout Widget',
    debugShowCheckedModeBanner: false,
    theme: ThemeData(
      primarySwatch: Colors.blue,
    ),
    home: MyHomePage(),
  ); } }
class MyHomePage extends StatelessWidget
{ @override
  Widget build(BuildContext context)
  { return Scaffold(
    backgroundColor: Colors.white,
    body: Center(
```

```
child: Stack(  
  alignment: Alignment.center,  
  children: <Widget>[  
    // Blue container at the bottom  
    Container(  
      color: Colors.blue,  
      height: 300.0,  
      width: 300.0,  
    ),  
    // Pink container in the middle  
    Container(  
      color: Colors.pink,  
      height: 200.0,  
      width: 200.0,  
    ),  
    // Yellow container on top  
    Container(  
      color: Colors.yellow,  
      height: 100.0,  
      width: 100.0,  
    ), ], ), ); } }
```

Output:

Practical No. 4

Title: - Designing the mobile app to implement different Gestures.

Input:

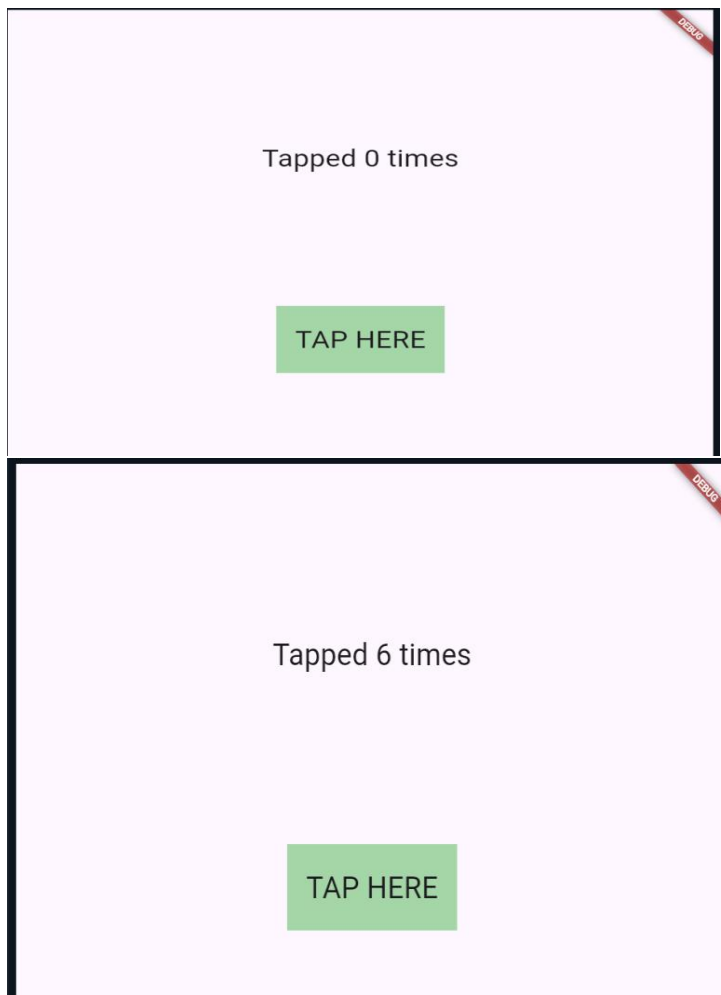
```
import 'package:flutter/material.dart';

void main()
{ runApp(MaterialApp(home:
MyApp()));
}

class MyApp extends StatefulWidget {
  const MyApp({Key? key}) : super(key: key);
  @override
  State<MyApp> createState() => _MyAppState();
}

class _MyAppState extends State<MyApp>
{ int numberOfTimesTapped = 0;
  @override
  Widget build(BuildContext context)
  { return Scaffold(
    body:
      Center( child:
        Column(
          mainAxisAlignment: MainAxisAlignment.spaceEvenly,
          children: [
            Text(
              'Tapped ' + numberOfTimesTapped.toString() + ' times',
              style: TextStyle(fontSize: 30),
            ),
            GestureDetector(
              onTap: () {
                setState()
```

```
{ numberOfTimesTapped++;  
  
  } ); },  
  child: Container(  
    padding: EdgeInsets.all(20),  
    color: Colors.green[200], child:  
    Text(  
      'TAP HERE',  
      style: TextStyle(fontSize: 30),  
    ), ), ), ], ), ), ); } }
```



Practical No. 5

Title: - Designing the mobile app to implement the theming and styling.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp(
    home: MyApp(), // MaterialApp
  )); }

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context)
  { return Scaffold(
    appBar: AppBar(
      title: Text('Theming and Styling'), // AppBar
    ),
    body:
      Center( child:
        Column(
          mainAxisAlignment: MainAxisAlignment.spaceEvenly,
          children: [
            Image.network(
              'https://static.vecteezy.com/system/resources/previews/010/938/844/non_2x/tropical-purple-butterfly-illustration-beautiful-butterfly-vector.jpg',
              height: 350,
              width: 350,
            ), // Image.network
          ], ), ), ); } }
```

Output:**Input:**

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    { const appName = 'Custom Themes';
    return MaterialApp(
      title: appName,
      theme:
      ThemeData( useMate
      rial3: true,
      colorScheme:
```

```

        ColorScheme.fromSeed( seedColor:
        Colors.purple,
brightness: Brightness.dark,),
        textTheme: const
        TextTheme( displayLarge:
        TextStyle(
        fontSize: 72,
        fontWeight: FontWeight.bold,
        ), ), ),
        home: MyHomePage(title: appName),
    ); } }

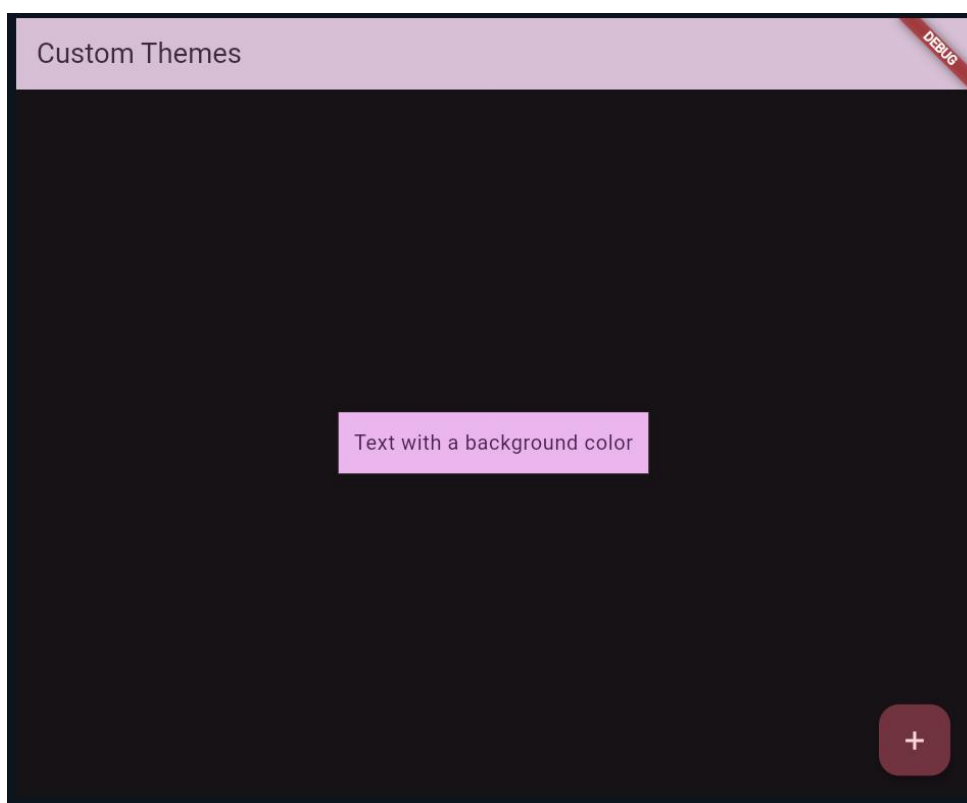
class MyHomePage extends StatelessWidget
{ final String title;
const MyHomePage({super.key, required this.title});
@override
Widget build(BuildContext context)
{ return Scaffold(
  appBar:
  AppBar( title:
  Text(
    title,
    style:
    Theme.of(context).textTheme.titleLarge!.copyWith( c
    olor: Theme.of(context).colorScheme.onSecondary,
    ), ),
  backgroundColor: Theme.of(context).colorScheme.secondary,
  ),
  body:
  Center( child:
  Container(
    padding: const EdgeInsets.symmetric(horizontal: 12, vertical: 12),

```

```

        color: Theme.of(context).colorScheme.primary,
        child: Text(
          'Text with a background color',
          style: Theme.of(context).textTheme.bodyLarge!.copyWith(
            color: Theme.of(context).colorScheme.onPrimary,
          ), ), ), ),
floatingActionButton: Theme(
  data:
    Theme.of(context).copyWith( colorSc
  heme:
    ColorScheme.fromSeed( seedColor:
    Colors.pink,
    brightness: Brightness.dark,
  ), ),
child:
  FloatingActionButton( onP
  ressed: () {},
  child: const Icon(Icons.add),
  ), ), ); } }

```

Output:

Practical No. 6

Title: - Designing the mobile app to create User Authentication.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context)
  { return const MaterialApp(
    title: 'User Authentication',
    home: Login(title: 'Login App'),
  ); } }

class Login extends StatefulWidget {
  const Login({Key? key, required this.title}) : super(key: key);
  final String title;

  @override
  State<Login> createState() => _LoginState();
}

class _LoginState extends State<Login>
{ final _formKey =
  GlobalKey<FormState>();
  TextEditingController emailController = TextEditingController();
  TextEditingController passwordController = TextEditingController();

  @override
  Widget build(BuildContext context) {
```

```

return
Scaffold( appBar:
AppBar(
  title: Text(widget.title),
),
body:
Form( key:
  _formKey,
child: Padding(
  padding: const EdgeInsets.symmetric(horizontal: 8, vertical: 16),
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.center,
    children: [
      Padding( paddin
        g:
          const EdgeInsets.symmetric(horizontal: 8, vertical: 16),
      child: TextFormField(
        controller: emailController,
        decoration: const
        InputDecoration( border:
        OutlineInputBorder(), labelText:
        "Email",
      ),
      validator: (value) {
        if (value == null || value.isEmpty)
          { return 'Please enter your email';
        }
        return null;
      }, ), ),
      Padding( paddin
        g:

```

```

        const EdgeInsets.symmetric(horizontal: 8, vertical: 16),
child: TextFormField(
  controller: passwordController,
  obscureText: true,
  decoration: const
    InputDecoration( border:
      OutlineInputBorder(), labelText:
        "Password",
    ),
  validator: (value) {
    if (value == null || value.isEmpty)
      { return 'Please enter your password';
    }
    return null;
  }, ), ),
Padding( paddin
g:
  const EdgeInsets.symmetric(horizontal: 8, vertical: 16.0),
child: Center(
  child:
    ElevatedButton( onPressed: () {
      if(_formKey.currentState!.validate()) {
        if(emailController.text == "abc@gmail.com" &&
          passwordController.text == "admin123")
          { Navigator.push(
            context,
            MaterialPageRoute(
              builder: (context) =>
                HomePage( email:
                  emailController.text,
```

```

        ), ), );
    } else
    { ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
        content: Text('Invalid Credentials'),
        ), ); }
    } else
    { ScaffoldMessenger.of(context).showSnackBar( c
    onst SnackBar(
        content: Text('Please fill input'),
        ), ); } },
    child: const Text('Submit'),
    ), ), ), ], ), ), ), ); } }
class HomePage extends StatelessWidget {
  const HomePage({super.key, required this.email});
  final String email;
  @override
  Widget build(BuildContext context)
  { return Scaffold(
    appBar: AppBar(
      title: const Text('Home Page'),
    ),
    body:
      Column( chil
      dren:
        [ Text(email),
        Center(
          child:
            ElevatedButton( onPressed: ()
              { Navigator.pop(context);
              },

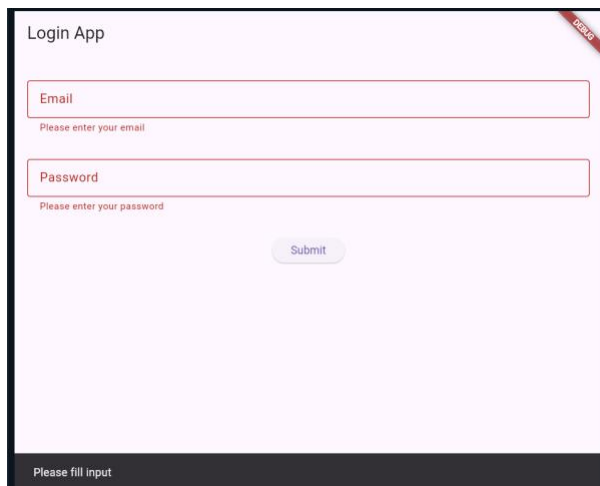
```



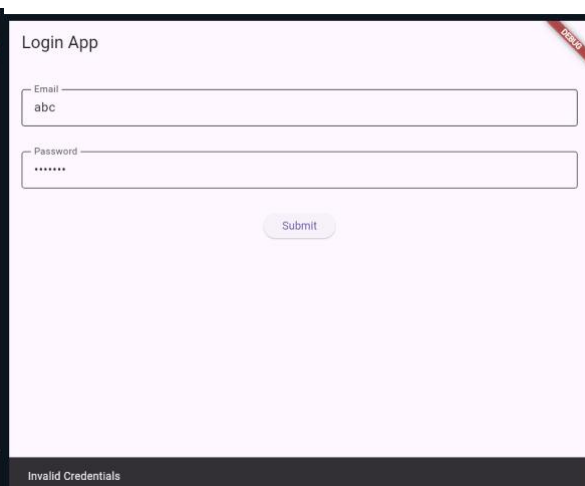
```
child: const Text("Go back!"),  
, ), ], ), ); } }
```

Output:

A screenshot of a mobile app titled "Login App". It features two input fields: "Email" and "Password", each with a placeholder text. Below the fields is a "Submit" button. A red "DEBUG" banner is visible in the top right corner.



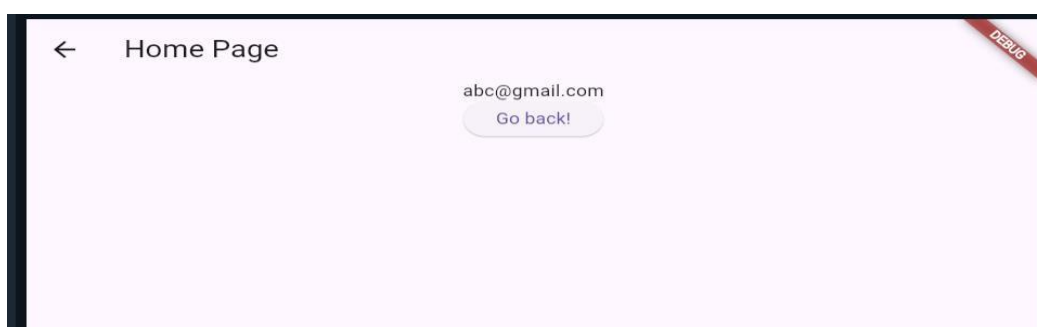
A screenshot of the "Login App" showing validation errors. The "Email" field has a red border and a message "Please enter your email" below it. The "Password" field also has a red border and a message "Please enter your password" below it. The "Submit" button is still present. A red "DEBUG" banner is in the top right corner. At the bottom, a dark bar contains the text "Please fill input".



A screenshot of the "Login App" showing an "Invalid Credentials" error. The "Email" field contains "abc" and the "Password" field contains "*****". The "Submit" button is present. A red "DEBUG" banner is in the top right corner. At the bottom, a dark bar contains the text "Invalid Credentials".



A screenshot of the "Login App" with valid credentials. The "Email" field contains "abc@gmail.com" and the "Password" field contains "*****". The "Submit" button is present. A red "DEBUG" banner is in the top right corner.



A screenshot of the "Home Page" after successful login. It features a back arrow icon, the text "Home Page", the email address "abc@gmail.com", and a "Go back!" button. A red "DEBUG" banner is in the top right corner.

Practical No. 7

Title: - Designing the mobile app to create Countdown Timer.

Input:

```
import 'package:flutter/material.dart';
import 'dart:async';

void main()
{ runApp(const
  MyApp());
}

class MyApp extends StatelessWidget
{ const MyApp({super.key});
  @override
  Widget build(BuildContext context)
  { return
    MaterialApp( debugShowCheckedMode
      Banner: false, title: 'Countdown Timer
      Example', theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const CodeDemo(),
    ); } }

class CodeDemo extends StatefulWidget
{ const CodeDemo({super.key});

  @override
  State<CodeDemo> createState() => _CodeDemoState();
}

class _CodeDemoState extends State<CodeDemo>
{ int _seconds = 0;
  int _minutes = 0;
```

```

int _hours = 0;
bool _isRunning = false;
Timer? _timer;
void _startTimer()
{
    setState(() {
        _isRunning = true;
    });
    _timer = Timer.periodic(const Duration(seconds: 1), (timer)
    {
        setState(() {
            if (_seconds > 0) {
                _seconds--;
            } else {
                if (_minutes > 0) {
                    _minutes--;
                    _seconds = 59;
                } else {
                    if (_hours > 0) {
                        _hours--;
                        _minutes = 59;
                        _seconds = 59;
                    } else {
                        _isRunning = false;
                        _timer?.cancel();
                    }
                }
            }
        });
    });
}

void _pauseTimer()
{
    setState(() {
        _isRunning = false;
    });
    _timer?.cancel();
}

```

```

void _cancelTimer()
{
  setState(() {
    _hours = 0;
    _minutes = 0;
    _seconds = 0;
    _isRunning = false;
  });
  _timer?.cancel();
}

@override
Widget build(BuildContext context)
{
  return Scaffold(
    appBar: AppBar(
      title: const Text('Countdown Timer'),
    ),
    body:
      SingleChildScrollView( padding:
        const EdgeInsets.all(16), child:
          Center(
            child: Column(
              mainAxisAlignment: MainAxisAlignment.start,
              children: [
                // Display time
                Container(
                  width: double.infinity,
                  height: 200,
                  color: Colors.amber,
                  child: Center(
                    child: Text(
                      '${_hours.toString().padLeft(2, '0')}:${_minutes.toString().padLeft(2, '0')}:${_seconds.toString().padLeft(2, '0')}',

```

```

        style: const TextStyle(
          fontSize: 60,
          fontWeight: FontWeight.bold,
        ), ), ), ),
const SizedBox(height: 50),
// Sliders for hours, minutes, seconds
Column(
  mainAxisAlignment: MainAxisAlignment.min,
  crossAxisAlignment: CrossAxisAlignment.center,
  children: [
    Text('Hours: $_hours', style: const TextStyle(fontSize: 20)),
    Slider(
      value: _hours.toDouble(),
      min: 0,
      max: 24,
      onChanged: (value)
        { if (!_isRunning)
          { setState(() {
            _hours = value.toInt();
          }); } },
      divisions: 100,
      label: 'Hours: $_hours',
    ),
    Text('Minutes: $_minutes', style: const TextStyle(fontSize: 20)),
    Slider(
      value: _minutes.toDouble(),
      min: 0,
      max: 59,
      onChanged: (value)
        { if (!_isRunning)
          { setState(() {

```

```

        _minutes = value.toInt();
    } ); } },
    divisions: 59,
    label: 'Minutes: $_minutes',
),
Text('Seconds: $_seconds', style: const TextStyle(fontSize: 20)),
Slider(
    value: _seconds.toDouble(),
    min: 0,
    max: 59,
    onChanged: (value)
    { if (!_isRunning)
    { setState(() {
        _seconds = value.toInt();
    } ); } },
    divisions: 59,
    label: 'Seconds: $_seconds',
), ], ),
const SizedBox(height: 50),
// Buttons
Row(
    mainAxisAlignment: MainAxisAlignment.spaceEvenly,
    children: [
        ElevatedButton( onPressed: () {
            if (_isRunning) {
                _pauseTimer();
            } else {
                _startTimer();
            } },

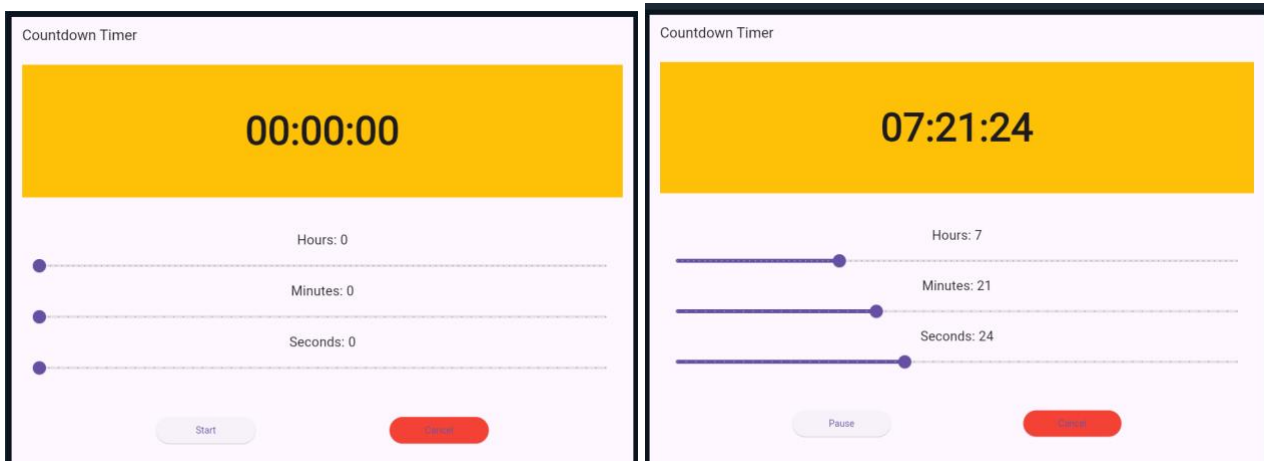
```

```

style:
  ElevatedButton.styleFrom( fixe
    dSize: const Size(150, 40),
  ),
  child: _isRunning ? const Text('Pause') : const Text('Start'),
),
ElevatedButton( onPressed:
  _cancelTimer,
  style:
    ElevatedButton.styleFrom( bac
      kggroundColor: Colors.red,
      fixedSize: const Size(150, 40),
    ),
    child: const Text('Cancel'),
  ), ], ), ], ), ), ), ); }

@override
void dispose() {
  if (_timer != null) {
    _timer!.cancel();
  }
  super.dispose();
} }

```

Output:

Practical no.8

Title: - Designing the mobile app to implement the routing.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp( h
    ome: MyApp(),
  )); }

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context)
  { TextEditingController name = TextEditingController();
    TextEditingController id = TextEditingController();
    TextEditingController semester = TextEditingController();
    TextEditingController dept = TextEditingController();
    TextEditingController city = TextEditingController();
    return Scaffold(
      appBar: AppBar(
        title: Text("Use Info"),
        centerTitle: true,
      ),
      body:
        Column( children:
          [ SizedBox(height:
            10), TextField(
              controller: name,
              decoration:
                InputDecoration( labelText:
                  "Enter your name",
```



```

        border: OutlineInputBorder( borderRadius:
            BorderRadius.circular(15),
        ), ), ),
    SizedBox(height: 10),
    TextField(
        controller: id,
        decoration:
            InputDecoration( labelText:
                "Enter your ID", border:
                    OutlineInputBorder(
                        borderRadius: BorderRadius.circular(15),
                    ), ), ),
    SizedBox(height: 10),
    TextField(
        controller: semester,
        decoration: InputDecoration(
            labelText: "Enter your Semester",
            border: OutlineInputBorder(
                borderRadius: BorderRadius.circular(15),
            ), ), ),
    SizedBox(height: 10),
    TextField(
        controller: dept,
        decoration:
            InputDecoration( labelText: "Enter
                your Department", border:
                    OutlineInputBorder(
                        borderRadius: BorderRadius.circular(15),
                    ), ), ),
    SizedBox(height: 10),
    TextField(

```

```

    controller: city,
    decoration:
      InputDecoration( labelText:
        "Enter your City", border:
          OutlineInputBorder(
            borderRadius: BorderRadius.circular(15),
          ), ), ),
    SizedBox(height: 10),
    ElevatedButton( onPressed: () {
      Navigator.push( context,
        MaterialPageRoute(
          builder: (context) =>
            NextScreen( name: name.text,
              id: id.text,
              semester: semester.text,
              dept: dept.text,
              city: city.text))).whenComplete(()
              { name.clear();
                id.clear();
                semester.clear();
                dept.clear();
                city.clear();
              } ); },
      child: Text("continue"),
    ), ], ), ); } }

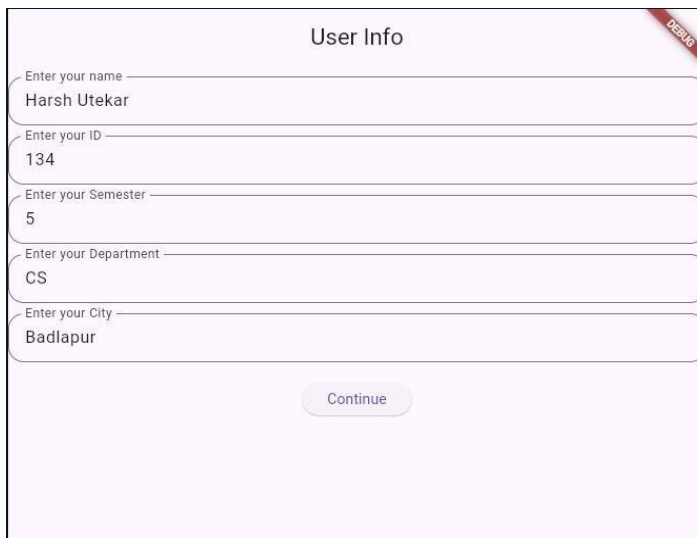
```

```

class NextScreen extends StatelessWidget
{ String? name, id, semester, dept, city;
  NextScreen(this.name, this.id, this.semester, this.dept, this.city);
  @override

```

```
Widget build(BuildContext context) {  
  return  
    Scaffold( body:  
      Column( child  
ren: [  
        Text("Name: " + name.toString()),  
        Text("Id: " + id.toString()),  
        Text("Semester: " + semester.toString()),  
        Text("Department: " + dept.toString()),  
        Text("City: " + city.toString()),  
      ], ), ); } }
```

Output:

The screenshot shows a mobile application interface with a light purple background. At the top, there is a title bar with the text "User Info". Below the title bar, there are five input fields, each with a label "Enter your [field name]" and a text input area. The input fields contain the following text: "Harsh Utekar", "134", "5", "CS", and "Badlapur". At the bottom of the form, there is a "Continue" button. A red "DEBUG" banner is visible in the top right corner of the app window.



The screenshot shows the output of the application, displaying the user information entered in the form. The text is as follows:
Name: Harsh Utekar
ID: 134
Semester: 5
Department: CS
City: Badlapur
A red "DEBUG" banner is visible in the top right corner of the app window.

Practical no.:- 9

Title: - Designing the mobile app to demonstrate navigation in an app.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      home: HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

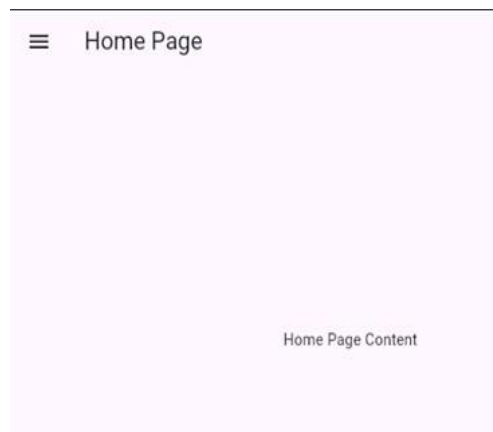
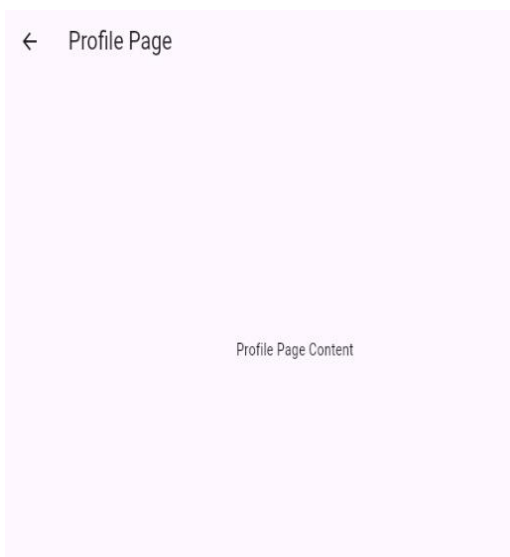
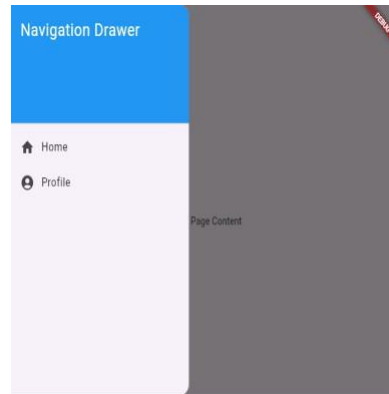
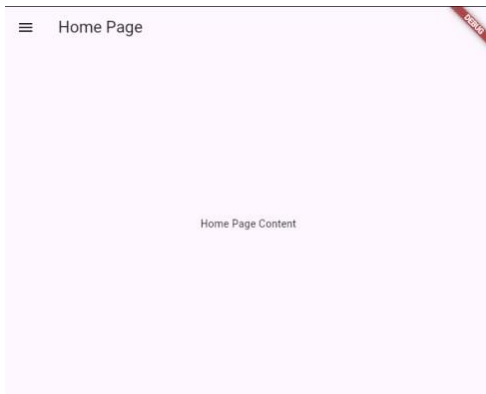
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Home Page')),
      drawer: Drawer(
        child: ListView(
          padding: EdgeInsets.zero,
          children:
            <Widget>[ const
            DrawerHeader(
              decoration:
                BoxDecoration( color:
```

```

        Colors.blue,
    ),
    child:
        Text( 'Navigation
        Drawer', style:
        TextStyle( color:
        Colors.white,
        fontSize: 24,
        ), ), ),
    ListTile(
        leading: const Icon(Icons.home),
        title: const Text('Home'),
        onTap: ()
        { Navigator.pop(context);
        }, ),
    ListTile(
        leading: const Icon(Icons.account_circle),
        title: const Text('Profile'),
        onTap: ()
        { Navigator.push( co
        ntext,
        MaterialPageRoute(builder: (context) => const ProfilePage()),
        ); }, ), ], ), ),
    body: const Center(child: Text('Home Page Content')),
  ); } }
class ProfilePage extends StatelessWidget
{ const ProfilePage({super.key});
@override
Widget build(BuildContext context)
{ return Scaffold(
  appBar: AppBar(title: const Text('Profile Page')), body:

```

```
const Center(child: Text('Profile Page Content'))  
); } }
```

Output:

Practical no.10

Title: - Designing the mobile app to create a BMI Calculator.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    { return const MaterialApp(
      debugShowCheckedModeBanner: false, // Remove debug banner
      title: 'BMI Calculator',
      home: BmiCalculatorScreen(),
    ); } }

class BmiCalculatorScreen extends StatefulWidget {
  const BmiCalculatorScreen({Key? key}) : super(key: key);

  @override
  State<BmiCalculatorScreen> createState() => _BmiCalculatorScreenState();
}

class _BmiCalculatorScreenState extends State<BmiCalculatorScreen> {
  { final TextEditingController _heightController =
    TextEditingController(); final TextEditingController _weightController =
    TextEditingController(); String _bmiResult = "";
    Color _resultColor = Colors.black;

  @override
  void dispose() {
    _heightController.dispose();
```

```
_weightController.dispose();
super.dispose();
}
void _calculateBMI() {
  final double? height = double.tryParse(_heightController.text);
  final double? weight = double.tryParse(_weightController.text);
  if (height == null || height <= 0 || weight == null || weight <= 0)
  { setState(() {
    _bmiResult = 'Please enter valid height and weight.';
    _resultColor = Colors.red;
  } );
  return;
}
final double bmi = weight / (height * height);
String category;
Color color;
if (bmi < 18.5) {
  category = 'Underweight';
  color = Colors.blue;
} else if (bmi >= 18.5 && bmi < 24.9)
{ category = 'Normal weight';
  color = Colors.green;
} else if (bmi >= 25 && bmi < 29.9)
{ category = 'Overweight';
  color = Colors.orange;
} else {
  category = 'Obesity';
  color = Colors.red;
}
setState(() {
```



```

    _bmiResult = 'BMI: ${bmi.toStringAsFixed(2)}\nCategory: $category';
    _resultColor = color;
  } ); }

```

```
@override
```

```
Widget build(BuildContext context)
```

```

{ return Scaffold(
  appBar: AppBar(
    title: const Text('BMI Calculator'),
    backgroundColor: Colors.deepOrange,
  ),
  backgroundColor: Colors.grey[100],
  body: Center(
    child:
      SingleChildScrollView( padding:
        const EdgeInsets.all(20), child:
          SizedBox(
            width: 380, // Slightly wider for better text fit on smaller screens
            child: Card(
              color: Colors.white,
              elevation: 10,
              shape:
                RoundedRectangleBorder( borderRadius:
                  BorderRadius.circular(15),
                ),
              child: Padding(
                padding: const EdgeInsets.all(25),
                child: Column(
                  mainAxisAlignment: MainAxisAlignment.min,
                  children: <Widget>[
                    const Text(
                      'Calculate Your Body Mass Index',

```

```
        style: TextStyle(
          fontSize: 20,
          fontWeight: FontWeight.bold,
          color: Colors.deepOrange,
        ),
        textAlign: TextAlign.center,
      ),
      const SizedBox(height: 25),
      TextField(
        controller: _heightController,
        keyboardType:
          const TextInputType.numberWithOptions(decimal: true),
        decoration: InputDecoration(
          labelText: 'Height (meters)',
          hintText: 'e.g., 1.75',
          border: OutlineInputBorder(
            borderRadius: BorderRadius.circular(10)),
          prefixIcon: const Icon(Icons.height),
        ),
      ),
      const SizedBox(height: 15),
      TextField(
        controller: _weightController,
        keyboardType:
          const TextInputType.numberWithOptions(decimal: true),
        decoration: InputDecoration(
          labelText: 'Weight (kilograms)',
          hintText: 'e.g., 70.5',
          border: OutlineInputBorder(
            borderRadius: BorderRadius.circular(10)),
          prefixIcon: const Icon(Icons.scale),
        ),
      ),
    ),
  ),
```

```

const SizedBox(height: 30),
ElevatedButton(
  onPressed: _calculateBMI,
  style:
    ElevatedButton.styleFrom( backgroundColor: Colors.deepOrange, padding: const
    EdgeInsets.symmetric(
      horizontal: 40, vertical: 15),
    shape: RoundedRectangleBorder(
      borderRadius: BorderRadius.circular(10),
    ), ),
  child: const
    Text( 'Calculate BMI',
      style: TextStyle(fontSize: 18, color: Colors.white),
    ), ),
const SizedBox(height: 25),
_bmiResult.isNotEmpty
  ? Text(
    _bmiResult,
    style:
      TextStyle( fontS
        ize: 22,
        fontWeight: FontWeight.bold,
        color: _resultColor,
      ),
    textAlign: TextAlign.center,
  )
: const SizedBox.shrink(),
], ), ), ), ), ), ); } }

```

Output:-

BMI Calculator

Calculate Your Body Mass Index

Height (meters)

Weight (kilograms)

Calculate BMI

BMI Calculator

Calculate Your Body Mass Index

Height (meters) 1.55

Weight (kilograms) 58

Calculate BMI

BMI: 24.14

Category: Normal weight

Practical no.11

Title: - Designing the mobile app to develop a simple TO-DO list app to manage the tasks.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return
      MaterialApp( debugShowCheckedMode
        Banner: false, home: TodoListScreen(),
      ); } }

class TodoListScreen extends StatefulWidget {
  const TodoListScreen({super.key});

  @override
  TodoListScreenState createState() => TodoListScreenState();
}

class TodoListScreenState extends State<TodoListScreen> {
  List<String> _todos = [];

  void _addTodo() {
    String newTodo =
      ";

    showDialog( contex
      t: context,
      builder: (BuildContext context) {
        { return AlertDialog(
```

```
title: const Text("Add a new To-Do"),
content:
```

```
  TextField( onChanged:
    (value) { newTodo =
      value;
    }, ),
```

```
actions:
```

```
  [ ElevatedButton(
    onPressed: () {
      setState() {
        if (newTodo.isNotEmpty) {
          _todos.add(newTodo);
        }
        Navigator.pop(context);
      } ); ],
    child: const Text("Add"),
  ), ], ); }, ); }
```

```
@override
```

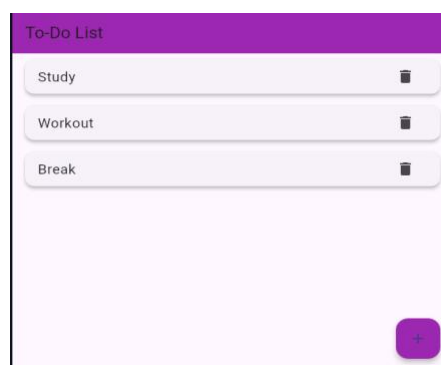
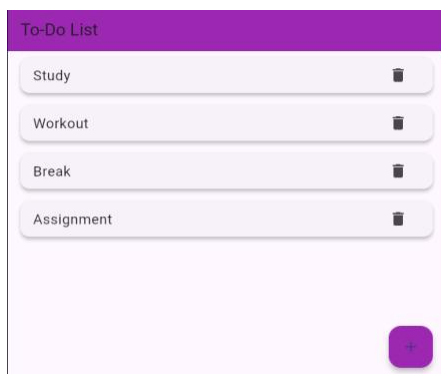
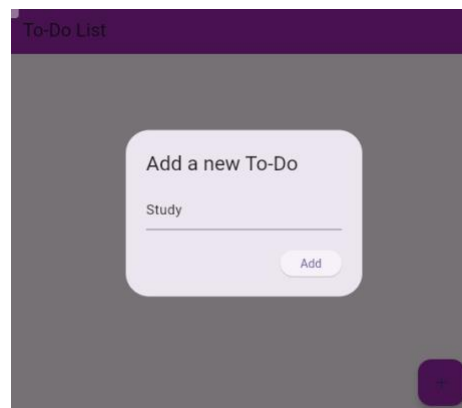
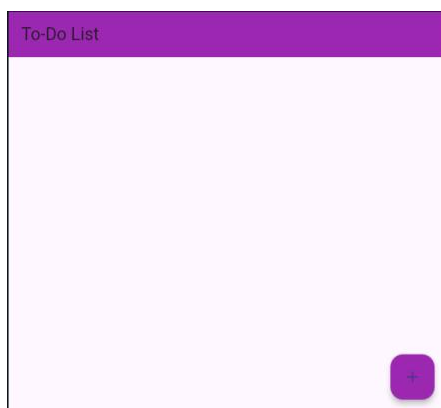
```
Widget build(BuildContext context)
```

```
{ return Scaffold(
  appBar: AppBar(
    title: const Text("To-Do List"),
    backgroundColor: Colors.purple,
  ),
  body:
    ListView.builder( itemCount:
      _todos.length, itemBuilder:
      (context, index) { return Card(
        elevation: 4,
        margin: const EdgeInsets.symmetric(vertical: 8, horizontal: 16),
        child: ListTile(
```

```

        title: Text(
          _todos[index],
          style: const TextStyle(fontSize: 18),
        ),
        trailing: IconButton(
          icon: const Icon(Icons.delete),
          onPressed: () {
            setState(() {
              _todos.removeAt(index);
            }); }, ), ), ); }, ),
floatingActionButton:
  FloatingActionButton( onPressed:
    _addTodo,
    backgroundColor: Colors.purple,
    child: const Icon(Icons.add),
  ), ); } }

```

Output:

Practical no.12

Title: - Designing the mobile app to implement an Image Slider and Image Gallery.

Input:

```
import 'package:flutter/material.dart';

void main()
{ runApp(ImageSliderApp());
}

class ImageSliderApp extends StatelessWidget
{ @override
Widget build(BuildContext context)
{ return
MaterialApp( debugShowCheckedMode
Banner: false, home: ImageSliderScreen(),
); } }

class ImageSliderScreen extends StatefulWidget
{ @override
_ImageSliderScreenState createState() => _ImageSliderScreenState();
}

class _ImageSliderScreenState extends State<ImageSliderScreen>
{ final PageController _pageController = PageController();
int _currentPage = 0;
final List<String> _images =
[ 'https://picsum.photos/id/1011/600/400',
'https://picsum.photos/id/1012/600/400',
'https://picsum.photos/id/1013/600/400',
'https://picsum.photos/id/1014/600/400',
'https://picsum.photos/id/1015/600/400',
];
```



```

@override
void initState()
{
  super.initState();
  _pageController.addListener()
  {
    setState(() {
      _currentPage = _pageController.page!.round();
    });
  }
}

@override
void dispose() {
  _pageController.dispose();
  super.dispose();
}

List<Widget> _buildPageIndicator() {
  return List.generate(_images.length, (index)
  {
    return Container(
      width: 10,
      height: 10,
      margin: EdgeInsets.symmetric(horizontal: 4),
      decoration: BoxDecoration(
        shape: BoxShape.circle,
        color: _currentPage == index ? Colors.blue : Colors.grey,
      ),
    );
  });
}

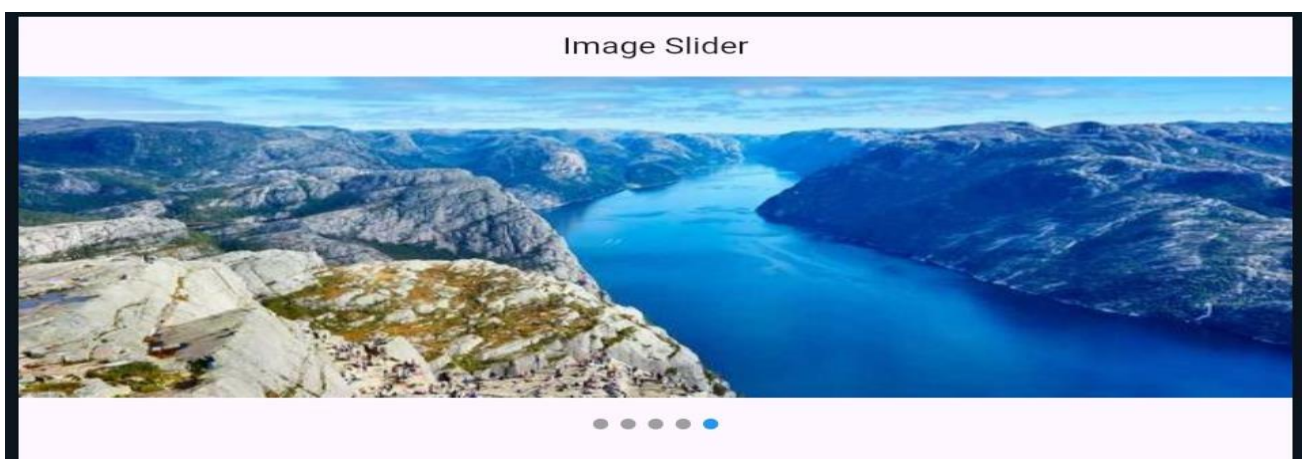
@override
Widget build(BuildContext context)
{
  return Scaffold(
    appBar: AppBar(
      title: Text('Image Slider'),
      centerTitle: true,
    ),
    body: Column(

```

children:

```
[ SizedBox(
  height: 300, // Prevents overflow
  child:
    PageView.builder( controller:
      _pageController, itemCount:
        _images.length, itemBuilder:
        (context, index) {
          return Image.network(
            _images[index],
            fit: BoxFit.cover,
            loadingBuilder: (context, child, loadingProgress)
              { if (loadingProgress == null) return child;
              return Center(child: CircularProgressIndicator());
            },
            errorBuilder: (context, error, stackTrace) {
              return Center(child: Text('Failed to load image'));
            }, ); }, ), ),
  SizedBox(height: 20),
  Row(
    mainAxisAlignment: MainAxisAlignment.center,
    children: _buildPageIndicator(),
  ), ], ), ); }
```

Output:



Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget
{
  @override
  Widget build(BuildContext context)
  {
    return
    MaterialApp( debugShowCheckedMode
    Banner: false, home:
    ImageGalleryScreen(),
    ); } }

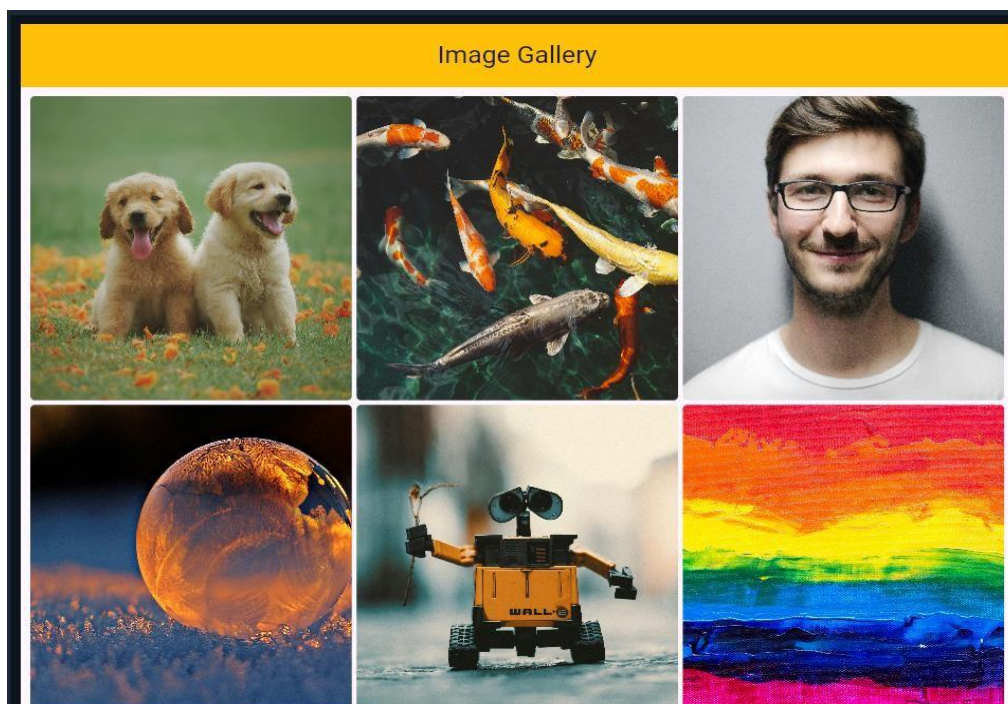
class ImageGalleryScreen extends StatelessWidget
{
  final List<String> _images = [
    'https://images.pexels.com/photos/1108099/pexels-photo-1108099.jpeg',
    'https://images.pexels.com/photos/213399/pexels-photo-213399.jpeg',
    'https://images.pexels.com/photos/220453/pexels-photo-220453.jpeg',
    'https://images.pexels.com/photos/302743/pexels-photo-302743.jpeg',
    'https://images.pexels.com/photos/247932/pexels-photo-247932.jpeg',
    'https://images.pexels.com/photos/1704120/pexels-photo-1704120.jpeg',
    'https://images.pexels.com/photos/34950/pexels-photo.jpg',
    'https://images.pexels.com/photos/733872/pexels-photo-733872.jpeg',
  ];

  @override
  Widget build(BuildContext context)
  {
    return Scaffold(
      appBar: AppBar(
        title: Text("Image Gallery"),
        centerTitle: true,
```

```

        backgroundColor: Colors.amber,
    ),
    body: Padding(
        padding: const EdgeInsets.all(8.0),
        child:
            GridView.builder( itemCount:
                _images.length,
                gridDelegate:
                    SliverGridDelegateWithFixedCrossAxisCount( crossAxisCount: 3,
                        crossAxisSpacing: 4,
                        mainAxisSpacing: 4,),
                itemBuilder: (context, index)
                { return ClipRRect(
                    borderRadius: BorderRadius.circular(4),
                    child: Image.network(
                        _images[index],
                        fit: BoxFit.cover,
                    ), ); }, ), ), ); } }

```

Output:

Practical no.13

Title: - Designing the mobile app to implement the state management.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    { return MaterialApp(
      debugShowCheckedModeBanner: false, // Hide the debug banner
      title: 'Checkbox Example',
      theme:
        ThemeData( useMat
          erial3: true,
        ),
      home: const HomePage(),
    ); } }

class HomePage extends StatefulWidget {
  const HomePage({Key? key}) : super(key: key);

  @override
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
  // Generate a list of available hobbies
  List<Map> availableHobbies = [
    {"name": "Football", "isChecked": false},
```

```

{"name": "Baseball", "isChecked": false},
{
  "name": "Video Games",
  "isChecked": false,
},
{"name": "Reading Books", "isChecked": false},
{"name": "Dancing", "isChecked": false}
];

@override
Widget build(BuildContext context)
{
  return Scaffold(
    appBar: AppBar(
      title: const Text('Checkbox Example'),
      backgroundColor: Colors.blueAccent,
      centerTitle: true,
    ),
    body:
      SingleChildScrollView( child:
        Padding(
          padding: const EdgeInsets.all(20),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              const Text(
                'Choose your hobbies:',
                style: TextStyle(fontSize: 24),
              ),
              const SizedBox(height: 10),
              const Divider(),
              const SizedBox(height: 10),
              // The checkboxes will be here
            ],
          ),
        ),
      ),
  );
}

```

```

Column(
  children: availableHobbies.map((hobby)
    { return CheckboxListTile(
      value: hobby["isChecked"],
      title: Text(hobby["name"]),
      onChanged: (newValue)
        { setState() {
          hobby["isChecked"] = newValue;
        } ); }, );
    } ).toList(),
),
// Display the result here
const SizedBox(height: 10),
const Divider(),
const SizedBox(height: 10),
Wrap(
  children: availableHobbies.map((hobby)
    { if (hobby["isChecked"] == true) {
      return
        Card( elevation: 3,
          color: Colors.amber,
          child: Padding(
            padding: const EdgeInsets.all(8.0),
            child: Text(hobby["name"]),
          ), ); }
    } ).toList(),
), ], ), ), ); } }

```

Output:

Checkbox Example

Choose your hobbies:

Football

☐

Baseball

☐

Video Games

☐

Reading Books

☐

Dancing

☐

Checkbox Example

Choose your hobbies:

Football

☐

Baseball

☒

Video Games

☐

Reading Books

☒

Dancing

☐

Baseball

Reading Books

Practical no.14

Title: - Designing the mobile app to implement an animation.

Input:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    { return const
    MaterialApp( debugShowCheckedMode
    Banner: false, title: 'Animation Example',
    home: HomePage(),
    ); } }

class HomePage extends StatefulWidget {
  const HomePage( {Key? key} ) : super(key: key);

  @override
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage>
  with TickerProviderStateMixin {
  bool _isPlay = false;
  late AnimationController _controller;

  @override
  void initState() {
    _controller =
    AnimationController( duration: const
    Duration(seconds: 1), vsync: this,
```

```

);
super.initState();
}
// Dispose the controller
@override
void dispose() {
  _controller.dispose();
  super.dispose();
}
@override
Widget build(BuildContext context)
{ return Scaffold(
  appBar: AppBar(
    title: const Text('Animated Play/Pause Button'),
    backgroundColor: Colors.amberAccent,
    centerTitle: true,
  ),
  body: Center(
    child:
      GestureDetector( onTap: () {
        if (_isPlay == false) {
          _controller.forward();
          _isPlay = true;
        } else {
          _controller.reverse();
          _isPlay = false;
        }
      },
    child: AnimatedIcon(
      icon: AnimatedIcons.play_pause,
      progress: _controller,

```

```
size: 200,  
color: Colors.blue,  
, ), ), ); } }
```

Output: